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Secure storage of Crypto Wallet Seed Phrase using ECC and splitting technique.

As cryptocurrency adoption grows security of wallet seed phrase become critical due to increasing number of attacks on crypto wallets. This paper proposes secure method for storing crypto wallet seed phrases using elliptic curve cryptography (ECC) and a splitting technique. Seed phrase is series of 12-24 words to recover a wallet exposure can result in complete loss of funds users store funds normally on paper or metal devices. These are vulnerable.

Solution proposed involves splitting seed phrase into 'n' parts where value of n is chosen by the user to adjust desired level of security. Each part is encrypted using ECC algorithm which offers strong encryption with smaller key sizes compared to RSA. encrypted parts can be stored in various physical/online locations. During recovery all parts must be correctly decrypted and combined to reconstruct original phrase

Method proposed is resilient to brute force and dictionary attacks due to ECC strength and randomness of key generation. Even if attacker compromises one (1) storage location without access to all parts and decryption keys, full phrase cannot be reconstructed. Performance evaluation is ongoing regarding of splitting or length of phrase.